Genomic & Pharmacogenomic Clinical Summary

Patient: Melinda Ehrle (Melinda_Ehrle) | **Date:** October 13, 2026

1. PRIORITY GENOMIC FINDINGS & CARRIER STATUS

GENE	VARIANT	ZYGOSITY	CLASSIFICATION	CLINICAL IMPACT & GUIDANCE
POLG	p.Gly737Arg · c.2209G>C · rs121918054	Heterozygous	Pathogenic	Strict contraindication against valproic acid due to fatal liver toxicity risk. Carrier retains normal baseline mitochondrial function.
TAT	p.Arg57Ter · c.169C>T · rs118203914	Heterozygous	Pathogenic	Single-copy heterozygous carrier; maintains normal enzymatic function on standard diet. No protein or tyrosine restriction required.
ATM	c.1236-2del · rs1565381646	Heterozygous	Pathogenic	Moderately elevated cancer susceptibility. Discuss guideline-directed breast/preventive screening (annual MRI/mammogram) with PCP.
VDR	c.-1172A>G · rs4516035	Heterozygous	Pathogenic	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
MC1R	p.Asp294His · c.880G>C · rs1805009	Heterozygous	Risk Factor	Increased UV skin sensitivity. Annual full-body skin examination and routine broad-spectrum SPF 30+ sun protection.
AMPD1	p.Glu435Gly · c.1304A>G · rs764950035	Heterozygous	VUS	Single-copy heterozygous carrier; normal muscle energy metabolism. Symptom-guided check only if exercise cramping or statin myalgia occurs.
CTH	p.Thr67Ile · c.200C>T · rs28941785	Heterozygous	Modifier	Preserves baseline homocysteine clearance. Review routine B-vitamin intake (B6, B12, folate) during annual wellness checks.

Prescribing Precautions & Annual Checkup Discussion Topics

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2. HIGH-PRIORITY MEDICATION & EXPOSURE PRECAUTIONS (CPIC / FDA)

GENE & DIPLOTYPE	PHENOTYPE	AFFECTED MEDICATIONS	PRESCRIBING ACTION & GUIDANCE
POLG p.Gly737Arg · c.2209G>C · rs121918054	Mitochondrial Carrier	⚠️ LIFE-THREATENING RISK **Valproic Acid / Sodium Valproate**	**ABSOLUTE CONTRAINDICATION (FATAL RISK):** High risk of fatal valproate-induced hepatic necrosis (FDA Boxed Warning / CPIC). Strictly avoid valproate formulations.
ABCG2 *ref/*rs2231142 (Q141K (rs2231142) G>T)	Decreased Function	DOSE_DOWN Allopurinol, Clozapine, Imatinib, Methotrexate	**DOSE_DOWN:** Allele G is associated with decreased dose of allopurinol or febuxostat in people with Gout as compared to allele A.
CYP2C19 *2/*2 (splice defect (rs12769205) A>G, splice defect (rs4244285) G>A)	Poor Metabolizer	⚠️ HIGH RISK · AVOID **Clopidogrel (Plavix)**	**AVOID / CONTRAINDICATION:** Significantly reduced active metabolite formation and high risk of cardiovascular ischemic events/stent thrombosis. Avoid clopidogrel; prescribe alternative antiplatelet agent (Prasugrel or Ticagrelor).
CYP3A43 *2/*3 (Y25fs (rs61469810) delA, P340A (rs680055) C>G)	Unknown	MONITOR CPIC Substrates	**MONITOR:** Titrate dosing per CPIC.
CYP3A5 *3/*3 (splice defect (rs776746) T>C)	Poor Metabolizer	STANDARD Tacrolimus (Prograf)	**STANDARD:** CYP3A5 *3/*3 non-expressor (Poor Metabolizer). Requires standard lower initial dosing (0.10-0.15 mg/kg/day) compared to CYP3A5 expressors to achieve target trough concentrations. Monitor blood levels (TDM).

3. HIGH-LEVEL PREVENTIVE CARE & ANNUAL CHECKUP TOPICS

PREVENTIVE CARE TOPIC	TRIGGERING GENE & VARIANT	TIMING	CLINICAL CONTEXT & GUIDANCE
EHR Valproate Allergy/Avoidance Flag & Routine Liver Panel Permanent Prescribing Precaution	POLG p.Gly737Arg · c.2209G>C · rs121918054 Heterozygous (Autosomal Dominant)	Permanent EHR Flag & Annual Checkup	Strict contraindication against valproic acid due to fatal liver toxicity risk. Carrier retains normal baseline mitochondrial function.
Reassurance Only — Normal Unrestricted Diet Optional Routine Metabolic Panel at Annual Checkup	TAT p.Arg57Ter · c.169C>T · rs118203914 Heterozygous (Autosomal recessive)	Routine Annual Checkup	Single-copy heterozygous carrier; maintains normal enzymatic function on standard diet. No protein or tyrosine restriction required.
Guideline-Directed Breast & Cancer Screening Discussion (per NCCN) Preventive Screening Review	ATM c.1236-2del · rs1565381646 Heterozygous (Autosomal recessive)	Annual Preventive Screening Review	Moderately elevated cancer susceptibility. Discuss guideline-directed breast/preventive screening (annual MRI/mammogram) with PCP.
Routine Vitamin D & Calcium Check Annual Wellness Panel	VDR c.-1172A>G · rs4516035 Heterozygous (Pathway Modifier)	Annual PCP Checkup	Modulates calcitriol receptor signaling. Check routine 25(OH)D at annual checkup to guide standard vitamin D3 supplementation.
Annual Skin Examination & Daily Sun Protection Dermatology / PCP Skin Check	MC1R p.Asp294His · c.880G>C · rs1805009 Heterozygous (Pathway Modifier)	Annual Preventive Skin Check	Increased UV skin sensitivity. Annual full-body skin examination and routine broad-spectrum SPF 30+ sun protection.

Clinical Perspective, Evidence Context & EHR Chart Summary

Patient: `Melinda Ehrle` (`Melinda_Ehrle`)

4. CLINICAL PERSPECTIVE: PROACTIVE CARE VS. AVOIDING OVER-INTERVENTION

✅ WHY THESE FINDINGS ARE USEFUL

- **Proactive Medication Safety:** Documenting pharmacogenomic diplotypes prevents avoidable drug side effects and subtherapeutic dosing before a prescription is started.
- **Routine Preventive Care:** High-level wellness topics integrate naturally into the annual primary care physical.
- **Specialist Coordination:** Sharing finding-specific precautions supports informed care without unnecessary medicalization.

🛑 WHY AGGRESSIVE INTERVENTION IS NOT NEEDED

- **Single-Copy Carrier Reassurance:** Single-copy recessive carriers retain a functional gene copy and maintain normal baseline health.
- **Absence of Biallelic Recessive Disease:** Zero two-copy pathogenic recessive disease genotypes are present in this profile.
- **Pathway Modifiers Are Not Diagnoses:** Common physiological modifiers should never trigger unprovoked medication changes or invasive testing.

5. CLINICAL EVIDENCE CONTEXT BY FINDING CATEGORY

FINDING CATEGORY	EVIDENCE TIER	CLINICAL INTERPRETATION BASIS
Primary Carrier & Susceptibility Findings (POLG, TAT, ATM)	High	Established clinical curation records (ClinVar / OMIM) with clear inheritance context.
Pharmacogenomic Diplotypes (ABCG2, CYP2C19, CYP3A43)	High	CPIC / DPWG clinical dosing guidelines with established star-allele assignments.
Physiological & Pathway Modifiers	Moderate	Population-level associations guiding general wellness discussions rather than diagnostic labeling.

6. COPY-PASTE PHYSICIAN HANDOFF / EHR CHART SUMMARY

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CLINICAL GENOMICS & PHARMACOGENOMICS SUMMARY (Melinda Ehrle | ID: Melinda_Ehrle):
- PRIORITY GENOMIC & CARRIER FINDINGS: POLG (p.Gly737Arg · c.2209G>C · rs121918054, Heterozygous); TAT (p.Arg57Ter · c.169C>T · rs118203914, Heterozygous); ATM (c.1236-2del · rs1565381646, Heterozygous); VDR (c.-1172A>G · rs4516035, Heterozygous); MC1R (p.Asp294His · c.880G>C · rs1805009, Heterozygous). Single-copy recessive carrier findings retain one functional gene copy and do not cause primary recessive disease.
- PHARMACOGENOMIC PRECAUTIONS (CPIC / FDA): ABCG2 *ref/*rs2231142. Review pharmacogenomic diplotypes prior to prescribing substrate medications.
- PREVENTIVE CARE PLAN: Discuss high-level wellness and screening topics (Section 3) during routine primary care checkups.
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